

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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Total Marks: 100

Examination Paper

Question Q1 (20 marks)

Consider the following Enhanced Entity-Relationship (EER) diagram:

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17)
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3)

Question Q2 (15 marks)

Consider a relation $R(\text{CustomerID}, \text{CustomerName}, \text{OrderID}, \text{OrderDate}, \text{ProductID}, \text{ProductName})$ with the following set of functional dependencies F over R : $F = \{\text{CustomerID} \rightarrow \text{CustomerName}, \text{OrderID} \rightarrow \text{OrderDate}, \text{ProductID} \rightarrow \text{ProductName}, \text{OrderID}, \text{ProductID} \rightarrow \text{CustomerID}, \text{OrderID}, \text{ProductID} \rightarrow \text{CustomerName}\}$. Analyze this relation for normalization to 3NF and discuss any potential anomalies.

- a) Draw the functional dependency diagram. (1)
- b) Use the attribute closure and identify the primary key for the given relation. (1)
- c) If we insert a row $\langle \text{John}, \text{New York}, \text{O123}, \text{2023-10-01}, \text{P456}, \text{Camera} \rangle$ into the given relation, what is the update anomaly that is violated? Give reasons. (2)
- d) Assume that the relation has a row, $\langle \text{Alice}, \text{Boston}, \text{O789}, \text{2023-09-15}, \text{P112}, \text{Laptop} \rangle$. If we remove the OrderID O789, what is the possible update anomaly that can be violated? Give reasons. (2)
- e) Using the above functional dependencies, design a set of 3NF relations for the above given relation. Show clearly each stage in deriving the 3NF relations. Also identify the normal form at each of these stages. (3)
- f) Do the update anomalies of part (c) and (d) hold in the 3NF relations? Discuss briefly. (2)
- g) Are the 3NF relations derived in part (e) still in 3NF? Justify your answer. (2)
- h) Are the 3NF relations derived in part (e) in BCNF? If not, bring the relations to BCNF. (2)

Question Q3 (25 marks)

A public library is developing a database system to manage its book inventory and patrons. The library aims to track books, their authors, and the members who borrow them. John is the senior database administrator responsible for designing and overseeing the database system's architecture. Alice and Bob are junior database administrators who assist John with the daily maintenance of the database. Alice is assigned to manage the books and authors database, while Bob oversees the patron records. Clara, a database developer reporting to Alice and Bob, is tasked with creating the database schema that defines the structure of books, authors, and patrons. Write a T-SQL statement to create a login for John with Windows authentication.

- a) Accept or refute the above statement justifying your answer. (2)
- b) Code Segment:
``sql
SELECT PatientID, Name, Age

```
FROM Patients
WHERE Age > 18
ORDER BY Name;
...
```

There are several different statements in the T-SQL language to retrieve data based on various needs. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used. (2)

c) Briefly explain the options available for transferring data between two tables within the library database. (2)

d) Briefly explain how authentication and authorization are achieved in Microsoft SQL Server. (4)

e) In the library's database system, Clara needs to have permissions to create new tables for book catalogs and updating records. Write a T-SQL statement to create a login for Clara with SQL Server authentication. (3)

f) Grant Clara necessary permissions to manage the books and authors tables without giving her full administrative rights. (3)

g) Assuming Alice's username is 'alice.admin', write T-SQL statements to assign her the responsibility of managing the authors database using a user-defined server role. (3)

h) Assuming Bob's login name is 'bob.user', write T-SQL statements to grant him permission to view and manage patron information. (3)

i) Assuming John's login name is 'john.senior', write T-SQL statements to allow him to perform all administrative tasks on the library database. (3)

Question Q4 (40 marks)

Consider the following relational database schema that stores information about university courses and student enrollments. The student relation captures details of students enrolled in various courses and the course relation contains information about courses offered by the university. The enrollment relation links students to the courses they are enrolled in.

student (sid, sname, sage, major)

course (cid, cname, credits)

enrollment (sid, cid, semester, grade)

a) Find the names of students who are enrolled in the course with course id 'C101'. (5)

b) Find the course names that have been taken by students who are majoring in

- 'Computer Science'. (5)
- c) Get the details of courses that have received an enrollment during the Fall 2023 semester. (5)
- d) Retrieve the names of students who have received an 'A' grade in any courses they are enrolled in. (5)
- e) How many students are enrolled in the course with course id 'C202'? (5)
- f) Express the above queries (a) to (d) in tuple calculus. (15)